Applications of skew monoidal categories

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Skew Monoidal Category

Definition

A (left) skew monoidal category is a category $\mathcal{C}$ equipped with a tensor $- \otimes - : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ and a unit $I \in \mathrm{ob}(\mathcal{C})$.

$$\begin{aligned} \alpha : (X \otimes Y) \otimes Z &\to X \otimes (Y \otimes Z) \\ \lambda : I \otimes X &\to X \\ \rho : X &\to X \otimes I \end{aligned}$$

$$\begin{tikzcd}
(I \otimes X) \otimes Y \arrow[rr, "\alpha_{I,X,Y}"] \arrow[dr, "\lambda_X \otimes 1_Y"'] & & I \otimes (X \otimes Y) \arrow[dl, "\lambda_{X \otimes Y}"] \\
& X \otimes Y &
\end{tikzcd}
\qquad
\begin{tikzcd}
(X \otimes I) \otimes Y \arrow[r, "\alpha_{X,I,Y}"] & X \otimes (I \otimes Y) \arrow[d, "1_X \otimes \lambda_Y"] \\
X \otimes Y \arrow[u, "\rho_X \otimes 1_Y"] \arrow[r, "1_{X \otimes Y}"'] & X \otimes Y
\end{tikzcd}
\qquad
\begin{tikzcd}
(X \otimes Y) \otimes I \arrow[rr, "\alpha_{I,X,Y}"] & & X \otimes (Y \otimes I) \\
& X \otimes Y \arrow[ul, "\rho_{X \otimes Y}"] \arrow[ur, "1_X \otimes \rho_Y"'] &
\end{tikzcd}$$

$$\begin{tikzcd}
I \arrow[rr, "1_I"] \arrow[dr, "\rho_I"'] & & I \\
& I \otimes I \arrow[ur, "\lambda_I"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
& (X \otimes Y) \otimes (Z \otimes W) \arrow[dr] & \\
(((X \otimes Y) \otimes Z) \otimes W) \arrow[ur] \arrow[d] & & X \otimes (Y \otimes (Z \otimes W)) \\
(X \otimes (Y \otimes Z)) \otimes W \arrow[rr] & & X \otimes ((Y \otimes Z) \otimes W) \arrow[u]
\end{tikzcd}$$

The skew sequent calculus

$$\frac{}{A \mid \longrightarrow A}\ \mathsf{id} \qquad \frac{A \mid \Gamma \longrightarrow C}{- \mid A, \Gamma \longrightarrow C}\ \mathsf{shift}$$

$$\frac{S \mid \Gamma \longrightarrow A \quad A \mid \Delta \longrightarrow C}{S \mid \Gamma, \Delta \longrightarrow C}\ \mathsf{scut} \qquad \frac{- \mid \Gamma \longrightarrow A \quad S \mid \Delta_0, A, \Delta_1 \longrightarrow C}{S \mid \Delta_0, \Gamma, \Delta_1 \longrightarrow C}\ \mathsf{ccut}$$

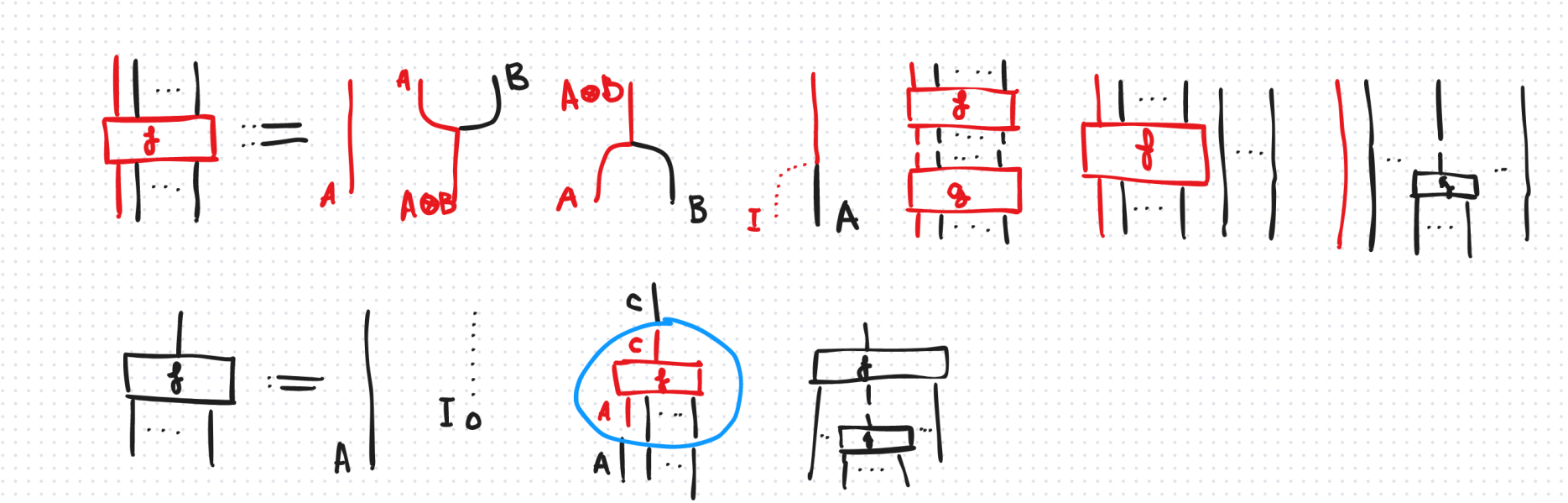
$$\frac{- \mid \Gamma \longrightarrow C}{\mathsf{I} \mid \Gamma \longrightarrow C}\ \mathsf{IL} \quad \frac{}{- \mid \longrightarrow \mathsf{I}}\ \mathsf{IR} \quad \frac{A \mid B, \Gamma \longrightarrow C}{A \otimes B \mid \Gamma \longrightarrow C}\ \otimes\mathsf{L} \quad \frac{S \mid \Gamma \longrightarrow A \quad - \mid \Delta \longrightarrow B}{S \mid \Gamma, \Delta \longrightarrow A \otimes B}\ \otimes\mathsf{R}$$

FIGURE 4. Rules of the skew monoidal sequent calculus

Theorem

The skew sequent calculus is sound and complete wrt the categorical calculus of a skew monoidal category.

A diagrammatic calculus for for skew monoidal categories



Remark

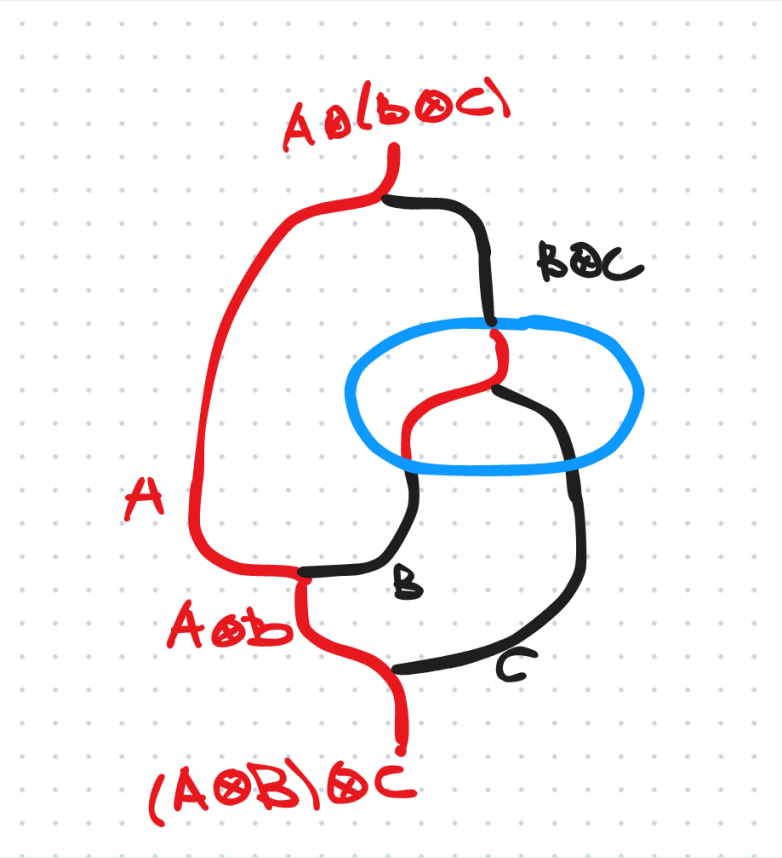
These are subject to certain equations.

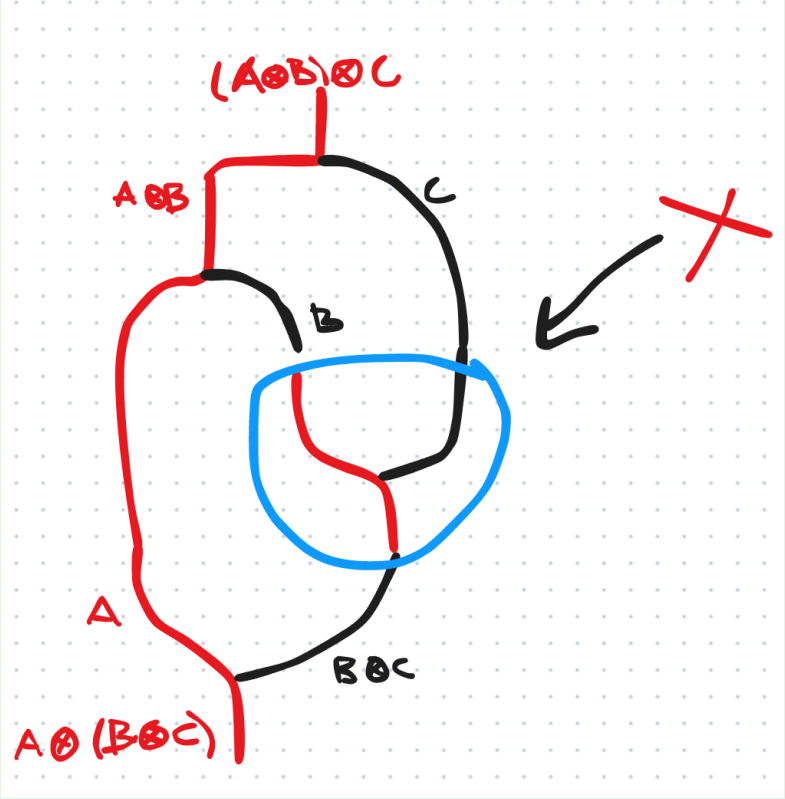
Definition

A diagram is *well-formed* using these constructors if it has one single output wire.

A diagrammatic calculus for for skew monoidal categories

Example (The associator as a diagram)

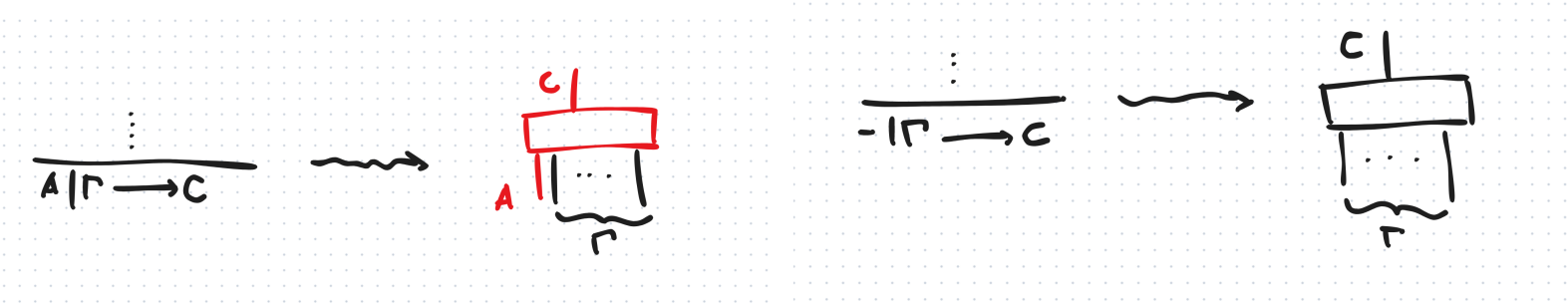


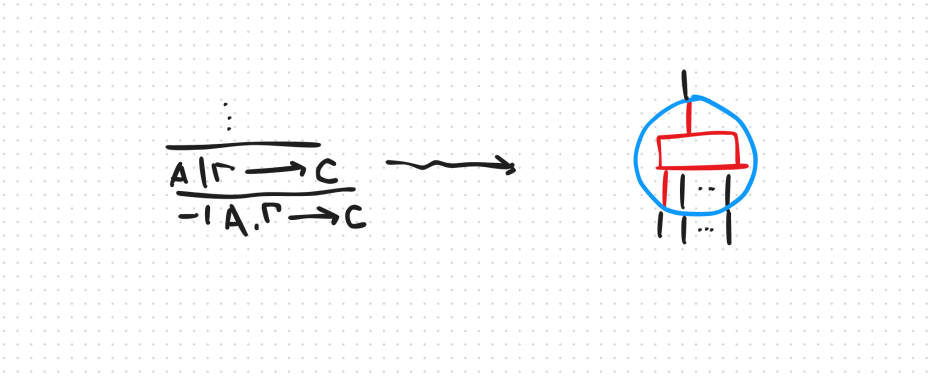


Explaination

Idea

- Red-wired diagrams - derivations concluded with stoup
- Black-wired diagrams - derivations concluded with empty stoup
- The blue "bubble" - *shift*-rule





Soundess and Completeness

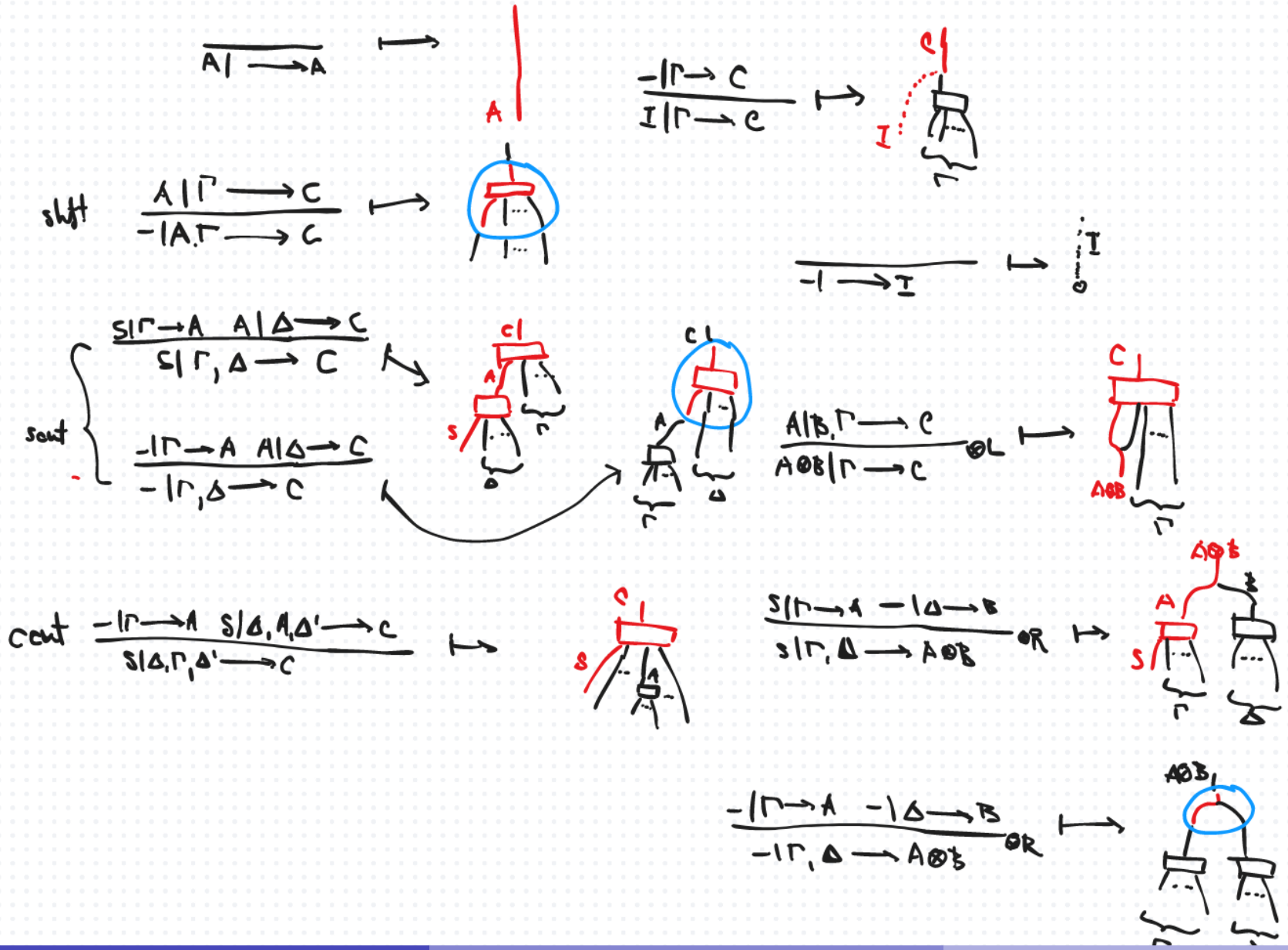
Conjecture?

The diagrammatic calculus is sound and complete wrt the skew sequent calculus.

Proof idea

We can define mutually inverse translations back-and-forth between the skew sequent calculus and the diagrammatic calculus.

Sequents to diagrams



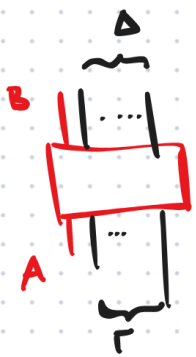
Diagrams to sequents

Problem

This direction is harder as generators do not necessarily correspond to well-formed diagrams!

Solution

We map in this way instead:



$$\longmapsto \quad \forall C\,((B \mid \Delta \to C) \to (A \mid \Gamma \to C))$$